

Hierarchical Loss Weighting for Physics-Informed Neural Networks in Small-Data Regimes

Abstract

Proxy models are widely employed in modern engineering to address problems with prohibitively high computational cost. Representative examples include uncertainty quantification tasks in petroleum engineering and CO₂ sequestration, where the direct approach requires a large number of numerical simulations of fluid flow in porous media. Such simulations are often infeasible in practice, particularly for high-resolution three-dimensional reservoir models.

A common strategy for reducing computational cost is to perform a limited number of fluid-flow simulations to generate training data for a proxy model. Owing to the expense of these simulations, the resulting datasets are typically small, which makes polynomial models, decision trees, and gradient-boosting methods the predominant choices for proxy modeling in this setting. In this work, we demonstrate that Physics-Informed Neural Networks (PINNs) are capable of changing this paradigm.

We conduct a series of numerical experiments showing that PINNs can be successfully trained on relatively small datasets (on the order of 500 samples). Furthermore, in scenarios where the target function exhibits strong nonlinearities, PINNs consistently outperform conventional proxy models, highlighting their potential as an effective and scalable alternative for data-limited, physics-driven engineering applications.

1 Introduction

Recent advances in machine learning have been largely driven by the availability of massive datasets. Enormous volumes of data are generated on a daily basis [Shirvani Moghaddam, 2024] and are routinely exploited to train state-of-the-art ML and AI models in domains such as social networks, online marketplaces, and streaming platforms. As a result, modern learning paradigms are often perceived as inherently data-hungry. However, many important real-world applications exist in which data acquisition is expen-

sive, time-consuming, or fundamentally limited. In such settings, the development of models that can be trained reliably on small datasets remains a problem of significant practical importance.

A representative example arises in petroleum engineering, where practitioners routinely address tasks such as sensitivity analysis (SA) and uncertainty quantification (UQ). These tasks rely on high-fidelity reservoir simulators that operate on fine spatial and temporal grids [Kennedy and O’Hagan, 2002] and account for multiple interacting physical effects [Zeng *et al.*, 2019]. While such models provide accurate predictions, they are computationally demanding, making large-scale SA and UQ prohibitively expensive. In practice, these analyses are often carried out using Monte Carlo or Markov Chain Monte Carlo methods [Cotter *et al.*, 2013; Lyon *et al.*, 2016; Beskos *et al.*, 2017; Giles, 2008], further amplifying the computational burden.

Due to the high cost of forward model evaluations, generating large datasets is typically infeasible. For instance, in uncertainty quantification studies for CO₂ sequestration [Dowell *et al.*, 2015], only on the order of 10³ simulation samples may be available. This characteristic has motivated the continued use of surrogate modeling techniques that are effective in small-data regimes. Polynomial approximation methods, such as Polynomial Chaos Expansion, have been successfully applied to a wide range of engineering problems. Tree-based models, including Decision Trees [Breiman *et al.*, 1984] and Gradient Boosted Decision Trees [Friedman, 2001; Chen and Guestrin, 2016], are also popular alternatives due to their robustness and favorable performance on limited tabular datasets.

In this work, we challenge the common perception of artificial neural networks (ANNs) as inherently data-demanding models. We demonstrate through a series of numerical experiments that ANNs trained within the Physics-Informed Neural Networks (PINNs) [Raissi *et al.*, 2019] framework can serve as accurate proxy models even when only small datasets are available. In simple settings, PINNs can be trained using solely the governing equations of the physical system, without relying on observational data. More generally, we show that in small-data regimes PINNs outperform classical surrogate models traditionally considered well-suited for limited datasets. Importantly, this performance gap widens as the complexity and nonlinearity of the target function increase.

These findings highlight the potential of PINNs as a viable and scalable proxy modeling tool for data-scarce engineering applications.

Physics-Informed Neural Networks approximate numerical solutions of differential equations by exploiting three complementary sources of information: (i) known solution values at a limited number of points, (ii) initial and boundary conditions, and (iii) the governing differential equations themselves. The discrepancy between PINN predictions and known solution values is typically quantified using standard regression losses such as the mean squared error (MSE). Enforcing consistency with initial and boundary conditions and with the governing equations requires computing derivatives of the network output with respect to its inputs. Thanks to automatic differentiation and backpropagation [Griewank and Walther, 2008; Rumelhart *et al.*, 1986], such derivatives can be computed efficiently using modern ML frameworks, including PyTorch [Paszke *et al.*, 2019]. In practice, the corresponding loss terms are evaluated by substituting the neural network approximation into the governing equations and measuring the magnitude of the resulting residuals.

Training a PINN thus requires the simultaneous minimization of multiple loss terms, which naturally leads to a multi-objective optimization problem. Many of the known challenges associated with PINNs stem from balancing these competing objectives. Consequently, a substantial body of research has focused on modifying the PINN training process, most commonly through weighted combinations of data, initial/boundary condition, and PDE residual losses. Selecting appropriate weights, however, remains an open problem, and numerous adaptive and dynamic weighting strategies have been proposed [Liu and Wang, 2021; Xiang *et al.*, 2022; Perez *et al.*, 2023]. Empirical studies have shown that PINNs may suffer from vanishing gradients, particularly in losses associated with initial and boundary conditions [Wang *et al.*, 2022]. To mitigate these issues, existing approaches often rely on gradient-based heuristics [Wang *et al.*, 2021] or on statistics of gradient magnitudes, such as norms or variances [Maddu *et al.*, 2022; Vemuri and Denzler, 2023; Deguchi and Asai, 2023].

An alternative line of research aims to improve PINN performance through adaptive sampling of collocation points [Wu *et al.*, 2023; Tang *et al.*, 2023; Daw *et al.*, 2023; Gao *et al.*, 2024] or by designing specialized activation functions tailored to PDE learning [Jagtap *et al.*, 2020; Gnanasambandam *et al.*, 2023]. These methods focus on improving either the expressiveness of the neural network or the efficiency with which the physical constraints are enforced.

In the present work, we propose a novel hierarchical dynamic loss-weighting strategy for PINNs. The key idea is to impose an explicit hierarchy among different information sources during training. At early stages, higher priority is given to data consistency, which is equivalent to a pre-training phase commonly used in deep learning. Once sufficient accuracy on the available data is achieved, the optimization focuses on satisfying initial and boundary conditions. Finally, after target accuracy levels for data and boundary losses are reached, the PDE residual loss becomes dominant, enforcing global physical consistency. This approach introduces only

two tunable hyperparameters corresponding to target values for the top levels of the hierarchy.

The intuition behind this strategy is that data and boundary constraints provide reliable local guidance for the optimizer, while PDE residual minimization propagates this information throughout the computational domain. Crucially, this hierarchical training procedure enables accurate PINN solutions even in small-data regimes, thereby expanding the applicability of PINNs to data-scarce engineering problems. Our main contributions can be summarized as follows:

- a novel hierarchical loss-weighting scheme for training Physics-Informed Neural Networks,
- extensive numerical evidence demonstrating its effectiveness on small datasets and highly nonlinear systems.

Accurate performance in small-data regimes should not be viewed as paradoxical. In PINNs, the governing equations and initial and boundary conditions encode complete physical knowledge of the system, effectively providing unlimited structural information. Sparse solution data serve primarily as guidance for the optimization process, helping the network avoid unfavorable local minima. From a data science perspective, physical constraints act as a powerful form of regularization, balancing fidelity to observed data with generalization across the domain.

2 Methodology

2.1 Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) are a class of neural network models that incorporate physical laws—typically expressed as ordinary or partial differential equations—directly into the training process. This integration enables PINNs to address both forward and inverse problems across a wide range of scientific and engineering applications, while ensuring that model predictions remain consistent with known physical principles.

A defining characteristic of PINNs is that learning does not rely exclusively on large labeled datasets. Instead, the governing equations, together with initial and boundary conditions, provide structural information that constrains the hypothesis space of the neural network. When observational or simulation data are available, they can be incorporated to guide training and to assess predictive accuracy. As a result, the formulation and balancing of the individual loss components play a crucial role in achieving stable and efficient optimization. In this work, we propose a hierarchical loss-weighting strategy designed to improve the convergence behavior of PINNs compared to the standard formulation with fixed loss weights.

2.2 Loss Function

Within the PINN framework, the solution of an ODE or PDE is approximated by an artificial neural network (ANN) that is required to satisfy three types of constraints: the governing differential equation, the initial and boundary conditions, and consistency with available data. In this work, we assume that the solution is known at a limited number of points, obtained from accurate classical numerical simulations. These

considerations naturally give rise to three complementary loss terms.

To enforce these constraints, we introduce a separate loss function for each component and combine them into a single weighted objective. Specifically, we consider the following losses:

- $\mathcal{L}_{\text{ibc}}$: loss enforcing initial and boundary conditions,
- $\mathcal{L}_{\text{pde}}$: loss penalizing violations of the governing differential equation,
- $\mathcal{L}_{\text{mse}}$: mean squared error (MSE) between PINN predictions and available data.

The overall training objective is defined as the weighted sum

$$\mathcal{L} = w_{\text{mse}}\mathcal{L}_{\text{mse}} + w_{\text{ibc}}\mathcal{L}_{\text{ibc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \quad (1)$$

where $w_{\text{mse}}, w_{\text{ibc}}, w_{\text{pde}}$ are non-negative weights normalized according to

$$w_{\text{mse}} + w_{\text{ibc}} + w_{\text{pde}} = 1. \quad (2)$$

Although PINNs can, in principle, be trained with arbitrary non-zero weights, in practice the choice of these weights has a significant impact on convergence speed and optimization stability. Poorly balanced loss contributions may result in slow convergence or entrapment in suboptimal local minima. To address this issue, we introduce a hierarchical loss-weighting strategy that dynamically adapts the contribution of each loss component during training.

Specifically, we introduce target values L_{mse}^0 , L_{ibc}^0 , and L_{pde}^0 for the respective loss terms. These targets are used to compute the corresponding weights as

$$\begin{cases} w_{\text{mse}} \propto 1, \\ w_{\text{ibc}} \propto \text{sg}\left(\exp\left(-\frac{\mathcal{L}_{\text{mse}}}{L_{\text{mse}}^0}\right)\right), \\ w_{\text{pde}} \propto \text{sg}\left(\exp\left(-\max\left(-\frac{\mathcal{L}_{\text{mse}}}{L_{\text{mse}}^0}, \frac{\mathcal{L}_{\text{ibc}}}{L_{\text{ibc}}^0}\right)\right)\right). \end{cases} \quad (3)$$

Here, $\text{sg}(\cdot)$ denotes the stop-gradient operator, ensuring that gradients of the loss weights are not propagated during backpropagation. Equations (2) and (3) uniquely determine the loss weights at each training iteration.

The intuition behind this weighting scheme is to enable a sequential utilization of different information sources during training. At early stages, the loss associated with observational data dominates, corresponding to a pre-training phase on simulation data. Once sufficient accuracy on the data points is achieved, the emphasis shifts toward enforcing initial and boundary conditions using randomly sampled boundary points. Finally, after boundary consistency is established, the PDE residual loss becomes dominant, shaping the global structure of the solution and enforcing physical consistency throughout the domain.

This hierarchical learning strategy can be summarized as follows:

- pre-training using available data,
- enforcement of initial and boundary conditions,
- global refinement through minimization of the PDE residual.

3 Numerical Examples

In this section, we evaluate the proposed hierarchical loss-weighting strategy on a set of benchmark problems of increasing complexity. First, we validate the method on a simple problem with a known analytical solution, enabling a quantitative assessment of approximation accuracy. Next, we consider a simplified two-dimensional oil reservoir displacement model to examine whether PINNs can be effectively trained in a small-data regime. Finally, we investigate a parametric extension of this model in which increased variability of physical parameters leads to highly nonlinear solution behavior. The objective of these experiments is to demonstrate that, under limited data availability, the proposed PINN framework yields accurate and robust approximations even in challenging regimes characterized by strong nonlinearity.

3.1 Heat Equation

As a first test case, we consider the one-dimensional heat equation with periodic boundary conditions. This problem describes the evolution of a temperature field over a spatial domain and admits a closed-form analytical solution, making it suitable for validating the proposed training strategy. The governing equations are given by

$$\begin{cases} \frac{\partial T}{\partial t} - \frac{\partial^2 T}{\partial x^2} = 0, \\ T(0, x) = 2 + \sin(2\pi x) + \cos(4\pi x), \\ T(t, 0) = T(t, 1), \end{cases} \quad (4)$$

where $T(t, x)$ denotes the temperature field. The solution is sought on the domain $x \in [0, 1]$ and $t \in [0, 0.05]$.

The exact solution of (4) is given by

$$T(t, x) = 2 + e^{-4\pi^2 t} \sin(2\pi x) + e^{-16\pi^2 t} \cos(4\pi x). \quad (5)$$

To approximate the solution, we employ a fully connected PINN that takes (x, t) as input and outputs the corresponding temperature value. The network consists of three hidden layers with 32 neurons, respectively, and uses the $\tanh$ activation function to mitigate vanishing gradients. A total of 100 collocation points are sampled uniformly from the spatio-temporal domain. Points corresponding to initial and boundary conditions are sampled independently along the associated manifolds. The model is trained for 20,000 epochs using full-batch optimization, which is feasible due to the moderate size of the training set.

The adaptive weighting scheme accelerates convergence and improves optimization stability by sequentially emphasizing data consistency, boundary adherence, and PDE residual minimization. This prevents premature dominance of individual loss components and leads to more reliable convergence behavior.

The proposed hierarchical loss-weighting strategy enables the PINN to accurately recover the analytical solution of (4). We compared our proposed solution with other algorithms that can be good predictors for small amount of data: XGBoost [Chen and Guestrin, 2016], Support Vector Machine Regression [Drucker *et al.*, 1997] and Regression with Gaussian Processes [Rasmussen and Williams, 2006]:

Metric	PINN	XGB	SVM	GP
MSE	$6.954 \cdot 10^{-3}$	$4.646 \cdot 10^{-2}$	$6.066 \cdot 10^{-2}$	$1.093 \cdot 10^{-2}$
RMSE	$8.339 \cdot 10^{-2}$	$2.155 \cdot 10^{-1}$	$2.463 \cdot 10^{-1}$	$1.045 \cdot 10^{-1}$
MAE	$5.392 \cdot 10^{-2}$	$9.020 \cdot 10^{-2}$	$1.440 \cdot 10^{-1}$	$5.667 \cdot 10^{-2}$
R^2	$9.589 \cdot 10^{-1}$	$7.253 \cdot 10^{-1}$	$6.413 \cdot 10^{-1}$	$9.354 \cdot 10^{-1}$

Table 1: Comparison of metrics on 2000 randomly sampled points. Best values are indicated in bold.

PINN with weighted strategy outperforms other methods. But GP achieves almost the same level of error. Others test will show that PINN will be better even than GP for more complicated tasks.

3.2 Two-Dimensional Two-Phase Displacement Problem

As a second test case, we consider a simplified two-dimensional model of waterflooding, a commonly used technique in oil recovery. The system describes the flow of two incompressible and immiscible fluids—water and oil—through a porous medium. For methodological clarity, we consider a square reservoir domain and focus on the development of a PINN-based surrogate model rather than on detailed geological realism.

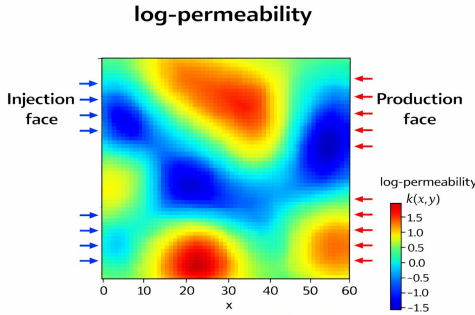


Figure 1: Permeability distribution

Water is injected along one boundary of the reservoir, displacing oil toward the opposite boundary, where fluid production occurs. The remaining two boundaries are assumed to be impermeable. The displacement process is characterized by the volumetric fractions (saturation) of water and oil, denoted by s_{water} and s_{oil} , respectively, which satisfy

$$s_{\text{water}} + s_{\text{oil}} = 1. \quad (6)$$

Fluid motion is governed by Darcy's law [Darcy, 1856]:

$$u_a(t, x) = - \frac{k(x) k_a(s(t, x))}{\mu_a} \nabla P(t, x), \quad (7)$$

where $a \in \{\text{water}, \text{oil}\}$, u_a denotes phase velocity, $k(x)$ is the permeability field, μ_a is fluid viscosity, and $k_a(\cdot)$ is the relative permeability. We employ the Corey parameterization [Corey, 1954]:

$$\begin{aligned} k_{\text{water}}(s) &= k_{\text{water}}^{(0)} s_{\text{water}}^{p_{\text{water}}}, \\ k_{\text{oil}}(s) &= k_{\text{oil}}^{(0)} s_{\text{oil}}^{p_{\text{oil}}}, \end{aligned} \quad (8)$$

with $k_{\text{water}}^{(0)} = 1$, $k_{\text{oil}}^{(0)} = 0.3$, $p_{\text{water}} = 2$, and $p_{\text{oil}} = 4$.

Mass conservation yields

$$\begin{cases} \phi \frac{\partial s_{\text{water}}}{\partial t} + \nabla \cdot u_{\text{water}} = 0, \\ \phi \frac{\partial s_{\text{oil}}}{\partial t} + \nabla \cdot u_{\text{oil}} = 0, \end{cases} \quad (9)$$

where ϕ denotes porosity.

We adopt dimensionless units and consider $x \in [0, 1]^2$, $t \in [0, 3]$, and pressure values in $[0, 1]$. Viscosities are set to $\mu_{\text{water}} = 1.0$ and $\mu_{\text{oil}} = 3.0$, and porosity is fixed at $\phi = 0.1$.

The initial conditions correspond to an oil-saturated reservoir:

$$s_{\text{water}}(0, x) = 0, \quad s_{\text{oil}}(0, x) = 1. \quad (10)$$

Boundary conditions include constant pressure and saturation at the injection boundary,

$$p(t, x)|_{x_1=0} = 1, \quad s_{\text{water}}(t, x)|_{x_1=0} = 1, \quad (11)$$

a fixed pressure at the production boundary,

$$p(t, x)|_{x_1=1} = 0, \quad (12)$$

and no-flow conditions on the remaining boundaries.

The permeability field is generated using a truncated Karhunen–Loève expansion [Ghanem and Spanos, 1991]. Two scenarios are considered, differing in the variance of the underlying random coefficients.

The model is trained using 100 spatio-temporal data points, full-batch optimization, and 10,000 epochs, resulting in a training time of approximately 5 minutes on a CPU.

We report predictive accuracy as a function of training set size and compare the proposed approach against classical surrogate models commonly used in small-data regimes as we listed in Section 2

The PINN is trained to predict pressure and phase saturations by minimizing residuals associated with mass conservation, Darcy's law, and the saturation constraint. The final loss function is constructed as a weighted sum of empirical residual means over domain, boundary, and data samples, following the methodology described in Subsection 3.1 and Table 1. There are following result for 2000 randomly sampled points (t, x, y) and simulation data of Pressure and Water saturation.

We report predictive accuracy as a function of training set size and compare the proposed approach against classical surrogate models commonly used in small-data regimes as we listed in Section 2.

Metric	PINN	XGB	SVM	GP
MSE	$4.624 \cdot 10^{-4}$	$1.303 \cdot 10^{-3}$	$5.986 \cdot 10^{-3}$	$2.126 \cdot 10^{-3}$
RMSE	$2.150 \cdot 10^{-2}$	$3.610 \cdot 10^{-2}$	$7.737 \cdot 10^{-2}$	$4.611 \cdot 10^{-2}$
MAE	$1.400 \cdot 10^{-2}$	$2.524 \cdot 10^{-2}$	$6.868 \cdot 10^{-2}$	$3.506 \cdot 10^{-2}$
R^2	$9.958 \cdot 10^{-1}$	$9.883 \cdot 10^{-1}$	$9.461 \cdot 10^{-1}$	$9.808 \cdot 10^{-1}$

Table 2: Metrics for pressure for small variance in permeability field.

The pressure is not hard to be predicted well, it doesn't change a lot with different domain points, hence metrics are almost the same, but still there is better prediction for PINN especially in MSE.

Metric	PINN	XGB	SVM	GP
MSE	$9.283 \cdot 10^{-4}$	$2.308 \cdot 10^{-3}$	$7.918 \cdot 10^{-3}$	$3.313 \cdot 10^{-3}$
RMSE	$3.047 \cdot 10^{-2}$	$4.804 \cdot 10^{-2}$	$8.898 \cdot 10^{-2}$	$5.756 \cdot 10^{-2}$
MAE	$1.025 \cdot 10^{-2}$	$1.670 \cdot 10^{-2}$	$7.285 \cdot 10^{-2}$	$3.415 \cdot 10^{-2}$
R^2	$8.767 \cdot 10^{-1}$	$6.934 \cdot 10^{-1}$	$5.741 \cdot 10^{-1}$	$5.599 \cdot 10^{-1}$

Table 3: Metrics for water saturation for small variance in permeability field.

In more complicated task - saturation predictions PINN performs much better in R^2 , this is crucial metric in physics systems because it emphasize how good model captures behavior of the system.

The next example will show more complicated example, where variance in permeability field has been increased three times. We get following results for Pressure and saturation:

Metric	PINN	XGB	SVM	GP
MSE	$1.325 \cdot 10^{-3}$	$1.286 \cdot 10^{-2}$	$1.255 \cdot 10^{-2}$	$1.691 \cdot 10^{-2}$
RMSE	$3.640 \cdot 10^{-2}$	$1.134 \cdot 10^{-1}$	$1.120 \cdot 10^{-1}$	$1.300 \cdot 10^{-1}$
MAE	$2.394 \cdot 10^{-2}$	$7.287 \cdot 10^{-2}$	$8.905 \cdot 10^{-2}$	$8.830 \cdot 10^{-2}$
R^2	$9.831 \cdot 10^{-1}$	$8.359 \cdot 10^{-1}$	$8.399 \cdot 10^{-1}$	$7.842 \cdot 10^{-1}$

Table 4: Metrics for pressure for high variance in permeability field.

Metric	PINN	XGB	SVM	GP
MSE	$8.288 \cdot 10^{-4}$	$1.053 \cdot 10^{-2}$	$9.808 \cdot 10^{-3}$	$1.105 \cdot 10^{-2}$
RMSE	$2.879 \cdot 10^{-2}$	$1.026 \cdot 10^{-1}$	$9.904 \cdot 10^{-2}$	$1.051 \cdot 10^{-1}$
MAE	$1.589 \cdot 10^{-2}$	$5.942 \cdot 10^{-2}$	$7.730 \cdot 10^{-2}$	$6.625 \cdot 10^{-2}$
R^2	$9.618 \cdot 10^{-1}$	$5.147 \cdot 10^{-1}$	$5.478 \cdot 10^{-1}$	$4.903 \cdot 10^{-1}$

Table 5: Metrics for water saturation for high variance in permeability field.

PINN continue to outperform other approaches, and with more complicated task mean squared error for SVM, XGBoost and GP increased 5-10 times, but for PINN it remains approximately the same.

We can see that error for PINN remains to be the lowest and order of error doesn't change much. But for other algorithms with higher task complexity error increasing and ability of models to provide accurate prediction become worse.

For PINN error order doesn't increasing, in some metrics it becomes even better. But for SVM, XGBoost and GP mean squared error becomes worse in 3-5 times.

3.3 Increased Dimensionality and High-Variance Permeability

In the final experiment, we assess the robustness of the proposed method in a highly nonlinear setting with parametric uncertainty. The model from Section 3.2 is extended by allowing key physical parameters to vary within the following ranges:

$$\begin{cases} k_{\text{water}}^{(0)} \in [0.9, 1.1], \\ k_{\text{oil}}^{(0)} \in [0.2, 0.4], \\ p_{\text{water}} \in [1.5, 2.5], \\ p_{\text{oil}} \in [3, 5]. \end{cases} \quad (13)$$

This variability induces strong nonlinear dependencies between inputs and outputs, making accurate approximation from limited data particularly challenging. The PINN is trained to predict pressure and instaurations at arbitrary spatial locations, time instances, and parameter values. As a result, the input space is seven-dimensional, while the output remains three-dimensional.

This experiment evaluates the ability of the proposed hierarchical loss-weighting strategy to maintain accuracy and stability in high-dimensional, data-scarce, and highly nonlinear regimes.

There are test metrics of pressure and water saturation for the models trained on 100 points:

Metric	PINN	XGB	SVM	GP
MSE	$7.300 \cdot 10^{-5}$	$9.500 \cdot 10^{-5}$	$2.707 \cdot 10^{-3}$	$2.248 \cdot 10^{-3}$
RMSE	$8.531 \cdot 10^{-3}$	$9.745 \cdot 10^{-3}$	$5.203 \cdot 10^{-2}$	$4.742 \cdot 10^{-2}$
MAE	$5.798 \cdot 10^{-3}$	$5.642 \cdot 10^{-3}$	$4.430 \cdot 10^{-2}$	$3.862 \cdot 10^{-2}$
R^2	$6.716 \cdot 10^{-1}$	$5.715 \cdot 10^{-1}$	$-1.122 \cdot 10^1$	$-9.146 \cdot 10^0$

Table 6: Metrics for pressure for high nonlinearity

Metric	PINN	XGB	SVM	GP
MSE	$8.190 \cdot 10^{-4}$	$5.197 \cdot 10^{-3}$	$1.262 \cdot 10^{-2}$	$1.101 \cdot 10^{-2}$
RMSE	$2.861 \cdot 10^{-2}$	$7.209 \cdot 10^{-2}$	$1.123 \cdot 10^{-1}$	$1.049 \cdot 10^{-1}$
MAE	$1.966 \cdot 10^{-2}$	$5.556 \cdot 10^{-2}$	$9.555 \cdot 10^{-2}$	$8.447 \cdot 10^{-2}$
R^2	$8.473 \cdot 10^{-1}$	$3.077 \cdot 10^{-2}$	$-1.354 \cdot 10^0$	$-1.053 \cdot 10^0$

Table 7: Metrics for water saturation for high nonlinearity

We show that even for very complicated tasks PINN outperforms other approaches on very limited training points. Because of high nonlinearity R^2 has decreased compare to previous tests, but still provide much better metrics. Moreover, SVM and GP become worsen then constant prediction for water saturation because of negative R^2 , hence they don't get behavior of physics system at all.

Moreover, despite high nonlinearity of the system, XGBoost showed negative R^2 for oil saturation prediction, it's values at $-3.750 \cdot 10^{-1}$, instead of it should provide approximately the same metrics for water and oil. PINN, SVM, GP show almost the same metrics for these characteristics. Therefore, it highlights that XGBoost can't describe behavior of this system here. With very limited number of points all the algorithms can't provide comparable approximation to PINN. But it should be mentioned that starting from 250 and 500 points XGBoost starting to provide better metrics for pressure and saturation. With $R^2 = 0.58$ and 0.61 , otherwise PINN still outperforms this algorithm. Moreover, starting from 500 points, both XGBoost and PINN showed R^2 greater than 90 for pressure with small MSE residual $8.190 \cdot 10^{-4}$ for PINN and $7.373 \cdot 10^{-3}$ for XGBoost. While SVM and GP show negative R^2 for pressure even if we increase number of training set in 20 times - to 2000 collocation points.

As we can state that prediction of pressure isn't very complicated task for PINN and XGBoost with more then 100 training points, let's clarify how will change MSE and R^2 of Water Saturation for every model if we will consider 100,

250, 500, 1000 and 2000 training points for every model. Because water saturation is more sparser then pressure, and therefore it's prediction become more complicated task.

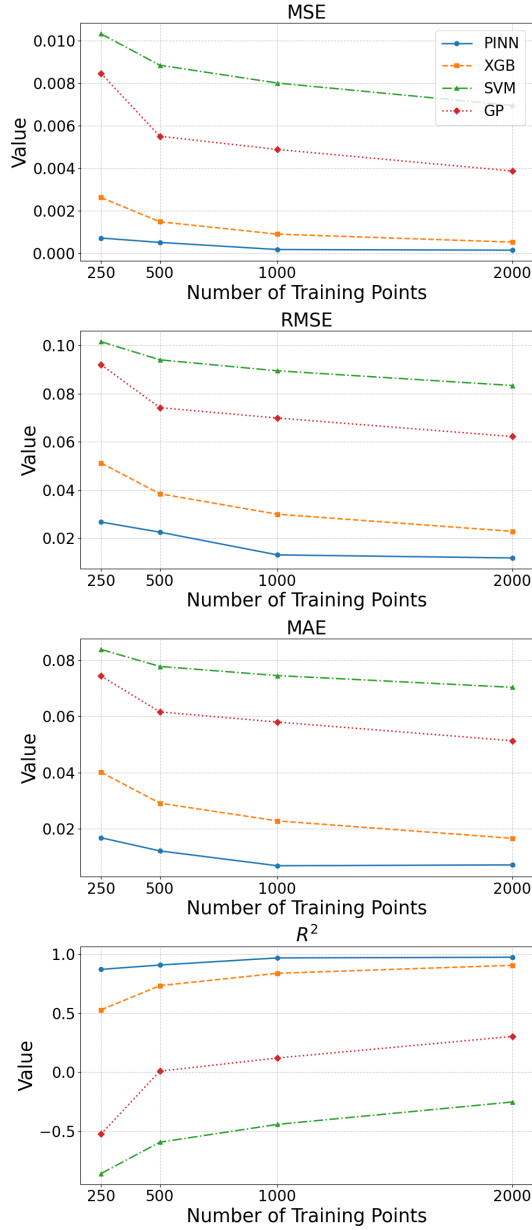


Figure 2: Dependency of water saturation metrics on the number of training points for the tested models.

It can be seen that PINN reach the best accuracy even with 8 times less points then other approaches. Moreover, starting with 250 training points PINNs provide very strong accuracy, and show that dependency between training dataset size and accuracy not that much as for other approaches. Regardless of the size of the training sample PINN provide very strong prediction.

4 Concluding Remarks

In this work, we demonstrated that lightweight Physics-Informed Neural Networks (PINNs) can serve as accurate proxy models for physical systems exhibiting strong nonlinearity or uncertainty, even in regimes where only small training datasets are available. The key enabler is a principled loss-weighting strategy that improves optimization stability by controlling the relative influence of data, initial/boundary conditions, and PDE residual losses during training.

Across all small-data experiments, the proposed approach yields consistently stronger saturation predictions than a competitive tree-based baseline (XGBoost), while maintaining comparable accuracy for pressure prediction, with differences typically within 5–6%. Importantly, in more challenging high-variance and highly nonlinear settings, PINNs retain strong predictive performance, whereas the baseline degrades substantially—particularly for saturation and, in the most extreme cases, also for pressure.

These results highlight the practical advantage of incorporating known physics directly into the learning objective. By leveraging governing equations and boundary/initial conditions as structural constraints, PINNs reduce reliance on dense labeled datasets and remain applicable when measurements are limited, expensive, or noisy. More broadly, our findings support the perspective that physics-based constraints act as an effective form of regularization, enabling data-efficient learning while preserving physical consistency.

As future work, we plan to extend the proposed loss-weighting strategy to three-dimensional subsurface flow models and to more realistic multi-physics settings. We also anticipate that combining hierarchical weighting with advances in adaptive sampling and hybrid physics–ML solvers will further expand the applicability of physics-informed learning to high-fidelity simulations under stringent data and computational constraints.

Ethical Statement

There are no ethical issues.

References

- [Beskos *et al.*, 2017] Alexandros Beskos, Ajay Jasra, Kody Law, Raul Tempone, and Yan Zhou. Multilevel sequential monte carlo samplers. *Stochastic Processes and their Applications*, 127(5):1417–1440, May 2017.
- [Breiman *et al.*, 1984] Leo Breiman, Jerome Friedman, Charles J Stone, and Richard A Olshen. *Classification and Regression Trees*. CRC press, 1984.
- [Chen and Guestrin, 2016] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pages 785–794, 2016.
- [Corey, 1954] Arthur T Corey. The interrelation between gas and oil relative permeabilities. *Producers Monthly*, 19(1):38–41, 1954.

- [Cotter *et al.*, 2013] S. L. Cotter, G. O. Roberts, A. M. Stuart, and D. White. Mcmc methods for functions: Modifying old algorithms to make them faster. *Statistical Science*, 28(3):424–446, 2013.
- [Darcy, 1856] Henry Darcy. *Les fontaines publiques de la ville de Dijon*. Dalmont, Paris, 1856.
- [Daw *et al.*, 2023] Arka Daw, Anuj P Marra, Mihai Anitescu, Brian Ziebart, et al. Mitigating propagation failures in physics-informed neural networks using retain-resample-release (r3) sampling. In *International Conference on Machine Learning*, pages 7265–7281. PMLR, 2023.
- [Deguchi and Asai, 2023] Shota Deguchi and Mitsuteru Asai. Dynamic & norm-based weights to normalize imbalance in back-propagated gradients of physics-informed neural networks. *Journal of Physics Communications*, 7(7):075005, 2023.
- [Dodwell *et al.*, 2015] T. J. Dodwell, C. Ketelsen, R. Scheichl, and A. L. Teckentrup. A hierarchical multilevel markov chain monte carlo algorithm with applications to uncertainty quantification in subsurface flow. *SIAM/ASA Journal on Uncertainty Quantification*, 3(1):1075–1108, 2015.
- [Drucker *et al.*, 1997] Harris Drucker, Christopher J C Burges, Linda Kaufman, Alex J Smola, and Vladimir Vapnik. Support vector regression machines. In *Advances in Neural Information Processing Systems*, volume 9, pages 155–161, 1997.
- [Friedman, 2001] Jerome H Friedman. Greedy function approximation: A gradient boosting machine. *Annals of statistics*, pages 1189–1232, 2001.
- [Gao *et al.*, 2024] Zhiwei Gao, Tao Tang, Liang Yan, and Tao Zhou. Failure-informed adaptive sampling for pinns, part ii: Combining with re-sampling and subset simulation. *Communications on Applied Mathematics and Computation*, 6(3):1720–1741, 2024.
- [Ghanem and Spanos, 1991] Roger G Ghanem and Pol D Spanos. *Stochastic Finite Elements: A Spectral Approach*. Springer, 1991.
- [Giles, 2008] Michael B. Giles. Multilevel Monte Carlo Path Simulation. *Operations Research*, 56(3):607–617, June 2008.
- [Gnanasambandam *et al.*, 2023] Raghav Gnanasambandam, Bo Shen, Jihoon Chung, Xubo Yue, and Zhenyu James Kong. Self-scalable tanh (stan): Multi-scale solutions for physics-informed neural networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(12):15588–15603, 2023.
- [Griewank and Walther, 2008] Andreas Griewank and Andrea Walther. *Evaluating derivatives: principles and techniques of algorithmic differentiation*. SIAM, 2 edition, 2008.
- [Jagtap *et al.*, 2020] Ameya D Jagtap, Kenji Kawaguchi, and George Em Karniadakis. Adaptive activation functions accelerate convergence in deep and physics-informed neural networks. *Journal of Computational Physics*, 404:109136, 2020.
- [Kennedy and O’Hagan, 2002] Marc C. Kennedy and Anthony O’Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 63(3):425–464, 01 2002.
- [Liu and Wang, 2021] Dehao Liu and Yan Wang. A dual-dimer method for training physics-informed neural networks. *Neural Networks*, 136:112–125, 2021.
- [Lyon *et al.*, 2016] R. J. Lyon, B. W. Stappers, S. Cooper, J. M. Brooke, and J. D. Knowles. Fifty years of pulsar candidate selection: from simple filters to a new principled real-time classification approach. *Monthly Notices of the Royal Astronomical Society*, 459(1):1104–1123, 04 2016.
- [Maddu *et al.*, 2022] Suryanarayana Maddu, Bevan L Sturm, Christian L Müller, and Ivo F Sbalzarini. Inverse dirichlet weighting enables reliable training of physics informed neural networks. *Machine Learning: Science and Technology*, 3(1):015026, 2022.
- [Paszke *et al.*, 2019] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035. Curran Associates, Inc., 2019.
- [Perez *et al.*, 2023] Sarah Perez, Suryanarayana Maddu, Ivo F Sbalzarini, and Philippe Poncet. Adaptive weighting of bayesian physics informed neural networks for multi-task and multiscale forward and inverse problems. *Journal of Computational Physics*, 491:112342, 2023.
- [Raissi *et al.*, 2019] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [Rasmussen and Williams, 2006] Carl Edward Rasmussen and Christopher K I Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [Rumelhart *et al.*, 1986] David E Rumelhart, Geoffrey E Hinton, and Ronald J Williams. Learning representations by back-propagating errors. *Nature*, 323(6088):533–536, 1986.
- [Shirvani Moghaddam, 2024] Shahriar Shirvani Moghaddam. The past, present, and future of the internet: A statistical, technical, and functional comparison of wired/wireless fixed/mobile internet. *Electronics*, 13:1986, 05 2024.
- [Tang *et al.*, 2023] Kejun Tang, Xiaoliang Wan, and Chao Yang. Das-pinns: A deep adaptive sampling method for solving high-dimensional partial differential equations. *Journal of Computational Physics*, 476:111868, 2023.

597 [Vemuri and Denzler, 2023] Sai Karthikeya Vemuri and
598 Joachim Denzler. Gradient statistics-based multi-objective
599 optimization in physics-informed neural networks. *Sen-*
600 *sors*, 23(21):8665, 2023.

601 [Wang *et al.*, 2021] Sifan Wang, Yujun Teng, and Paris
602 Perdikaris. Understanding and mitigating gradient
603 pathologies in physics-informed neural networks. *SIAM*
604 *Journal on Scientific Computing*, 43(5):A3055–A3081,
605 2021.

606 [Wang *et al.*, 2022] S. Wang, Y. Teng, and P. Perdikaris. Un-
607 derstanding and mitigating gradient flow pathologies in
608 physics-informed neural networks. *SIAM Journal on Sci-*
609 *entific Computing*, 44(1):A193–A219, 2022.

610 [Wu *et al.*, 2023] Chenxi Wu, Min Zhu, Qinyang Tan, Yadhu
611 Kartha, and Lu Lu. A comprehensive study of non-
612 adaptive and residual-based adaptive sampling for physics-
613 informed neural networks. *Computer Methods in Applied*
614 *Mechanics and Engineering*, 403:115671, 2023.

615 [Xiang *et al.*, 2022] Zixue Xiang, Wei Peng, Xiaodong
616 Zheng, Xinyu Zhao, and Wenyao Chen. Self-adaptive loss
617 balanced physics-informed neural networks. *Neurocom-*
618 *puting*, 496:11–34, 2022.

619 [Zeng *et al.*, 2019] Kaiyue Zeng, Jason Hou, Kostadin
620 Ivanov, and {Matthew Anderson} Jessee. Uncertainty
621 quantification and propagation of multiphysics simulation
622 of the pressurized water reactor core. *Nuclear Technology*,
623 205(12):1618–1637, December 2019. Publisher Copy-
624 right: © 2019, © 2019 American Nuclear Society.